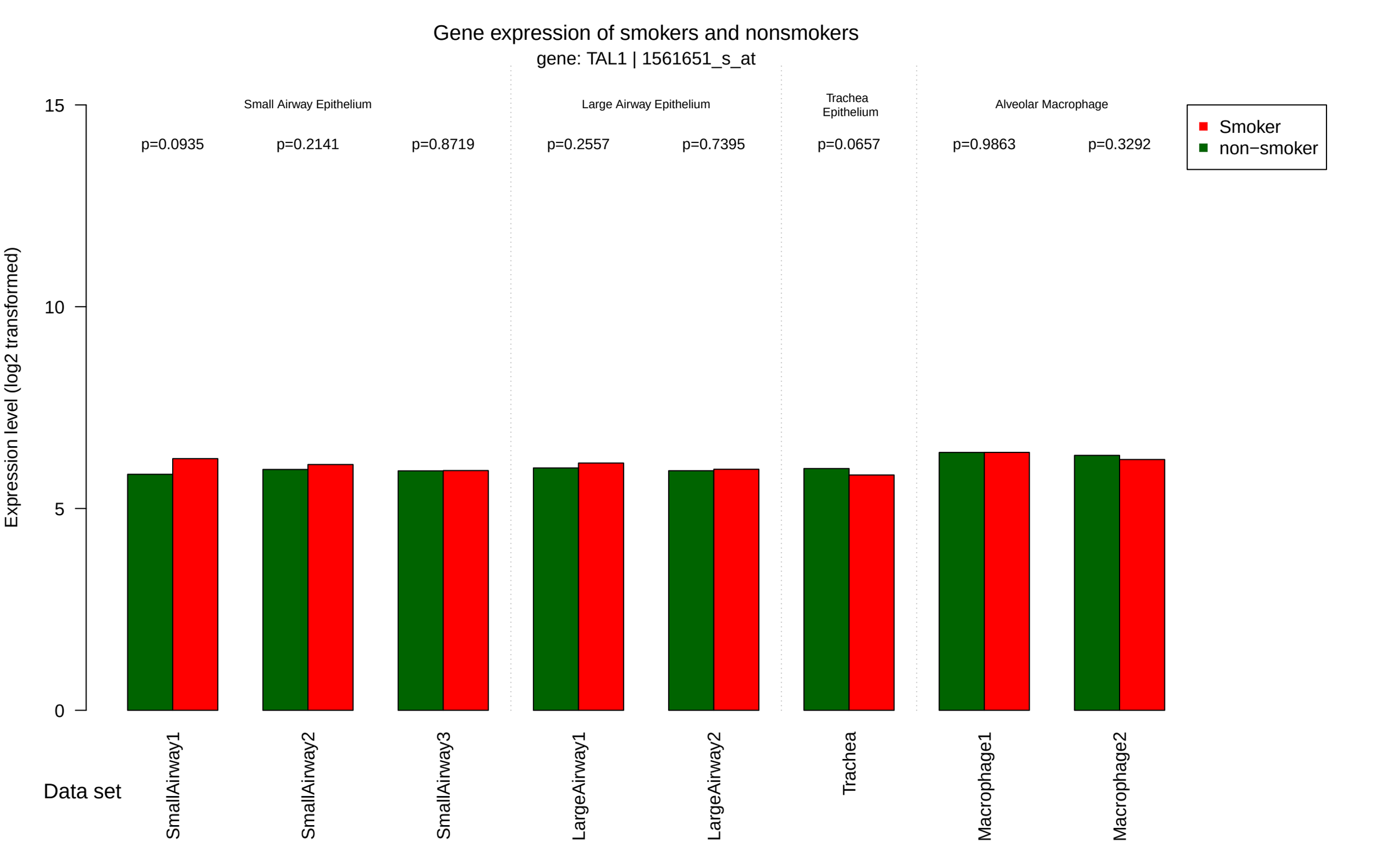
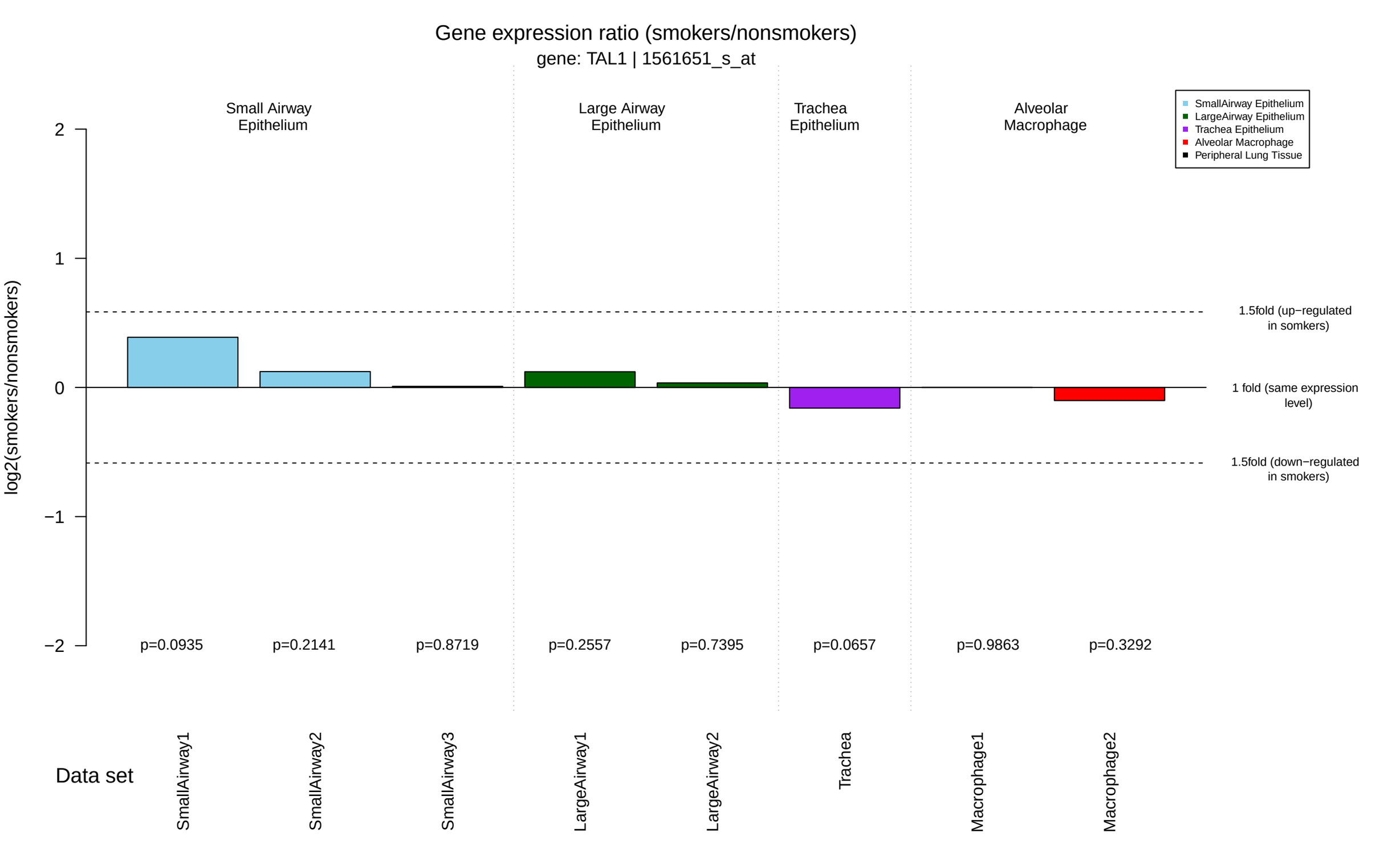
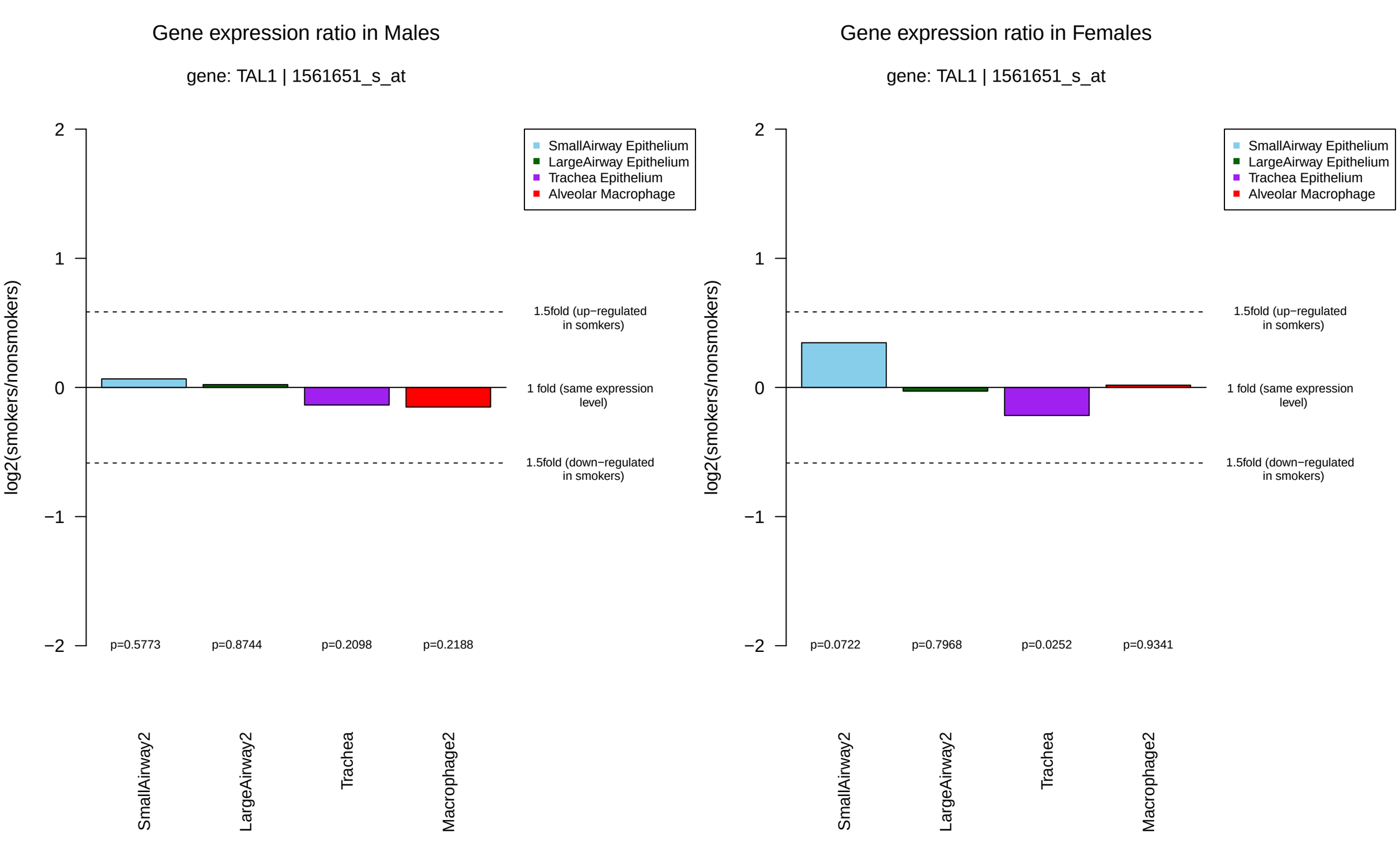
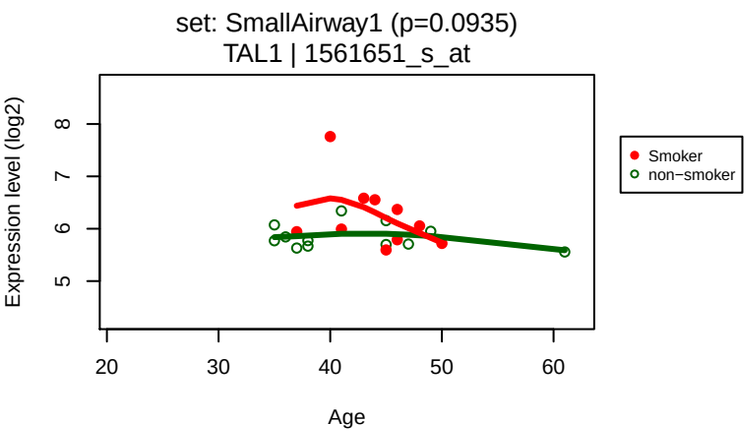
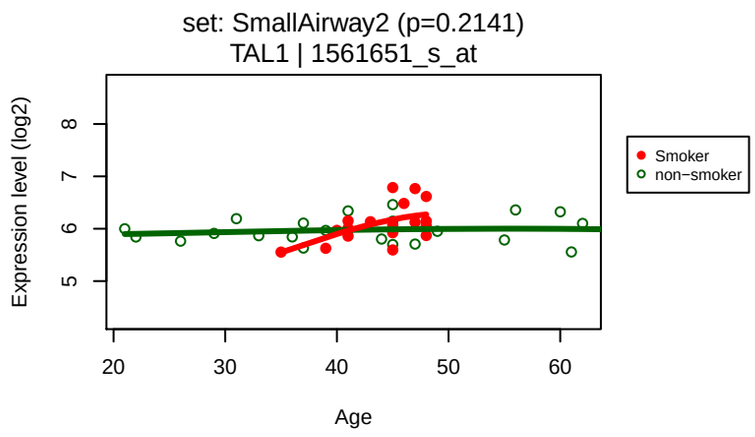
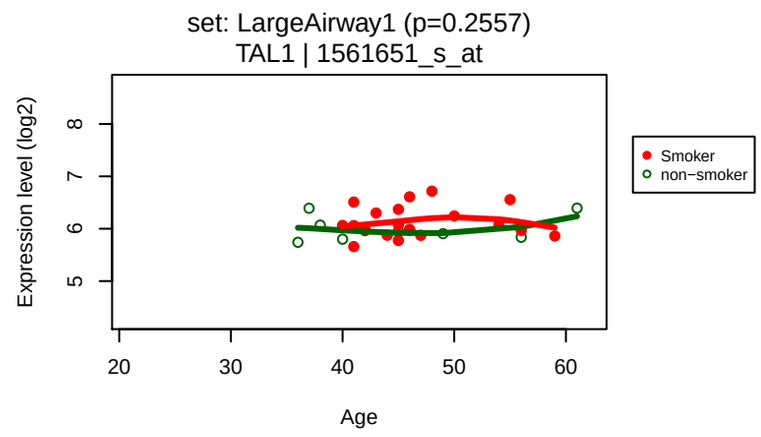

Gene expression of smokers and nonsmokers
gene: TAL1 | 1561651_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0935
p=0.2141
p=0.8719
p=0.2557
p=0.7395
p=0.0657
p=0.9863
p=0.3292
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

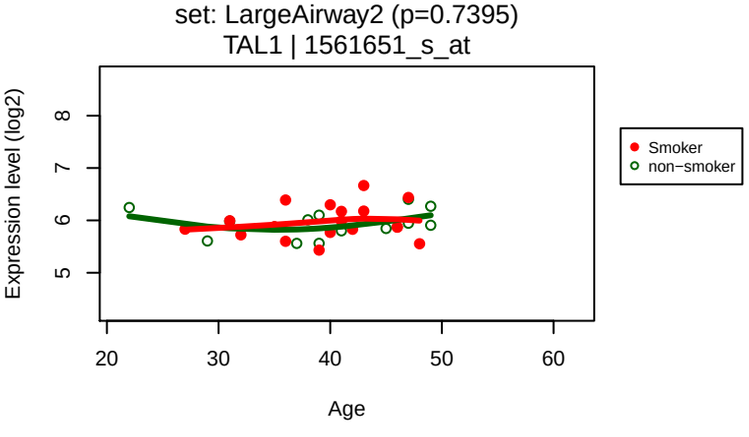

Gene expression ratio (smokers/nonsmokers)
gene: TAL1 | 1561651_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0935
p=0.2141
p=0.8719
p=0.2557
p=0.7395
p=0.0657
p=0.9863
p=0.3292
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

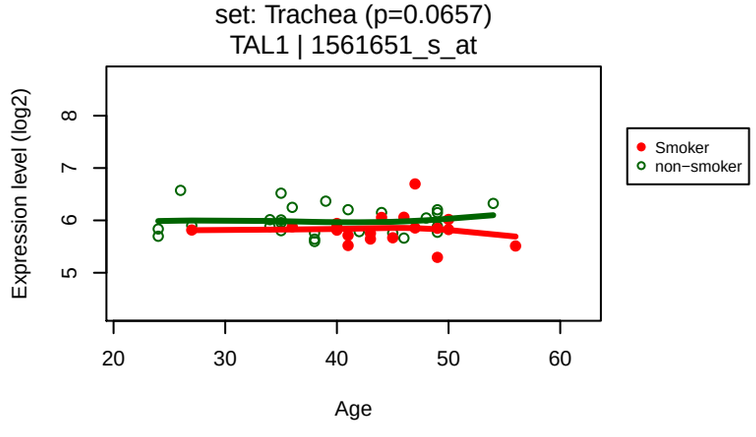

Gene expression ratio in Males
gene: TAL1 | 1561651_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5773
p=0.8744
p=0.2098
p=0.2188
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: TAL1 | 1561651_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0722
p=0.7968
p=0.0252
p=0.9341
SmallAirway2
LargeAirway2
Trachea
Macrophage2

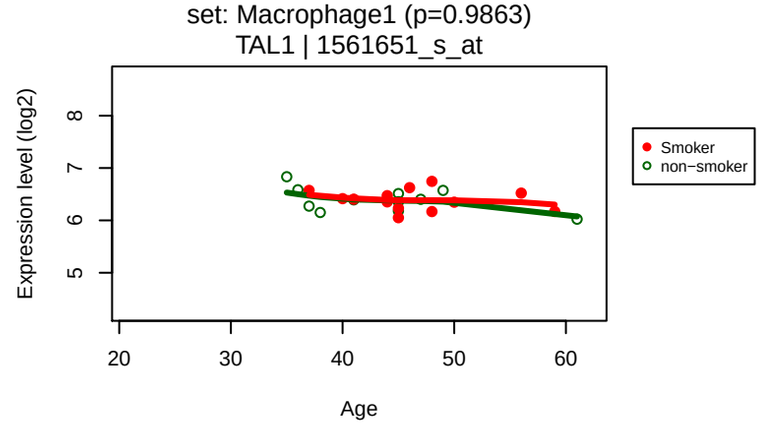

set: SmallAirway1 (p=0.0935)
TAL1 | 1561651_s_at
Expression level (log2)
Age
Smoker
non−smoker

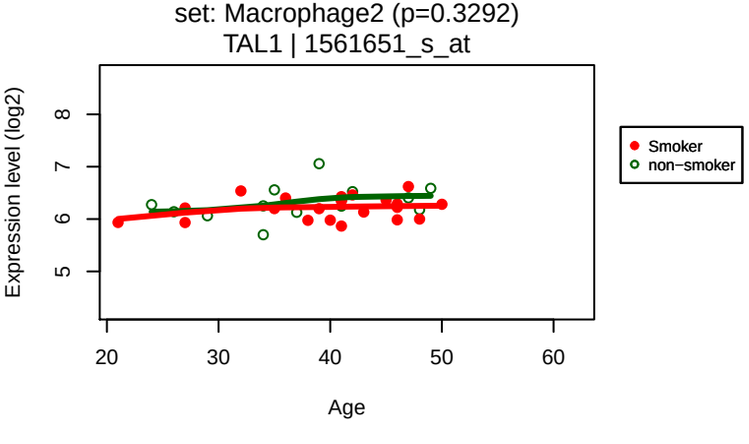

set: SmallAirway2 (p=0.2141)
TAL1 | 1561651_s_at
Expression level (log2)
Age
Smoker
non−smoker

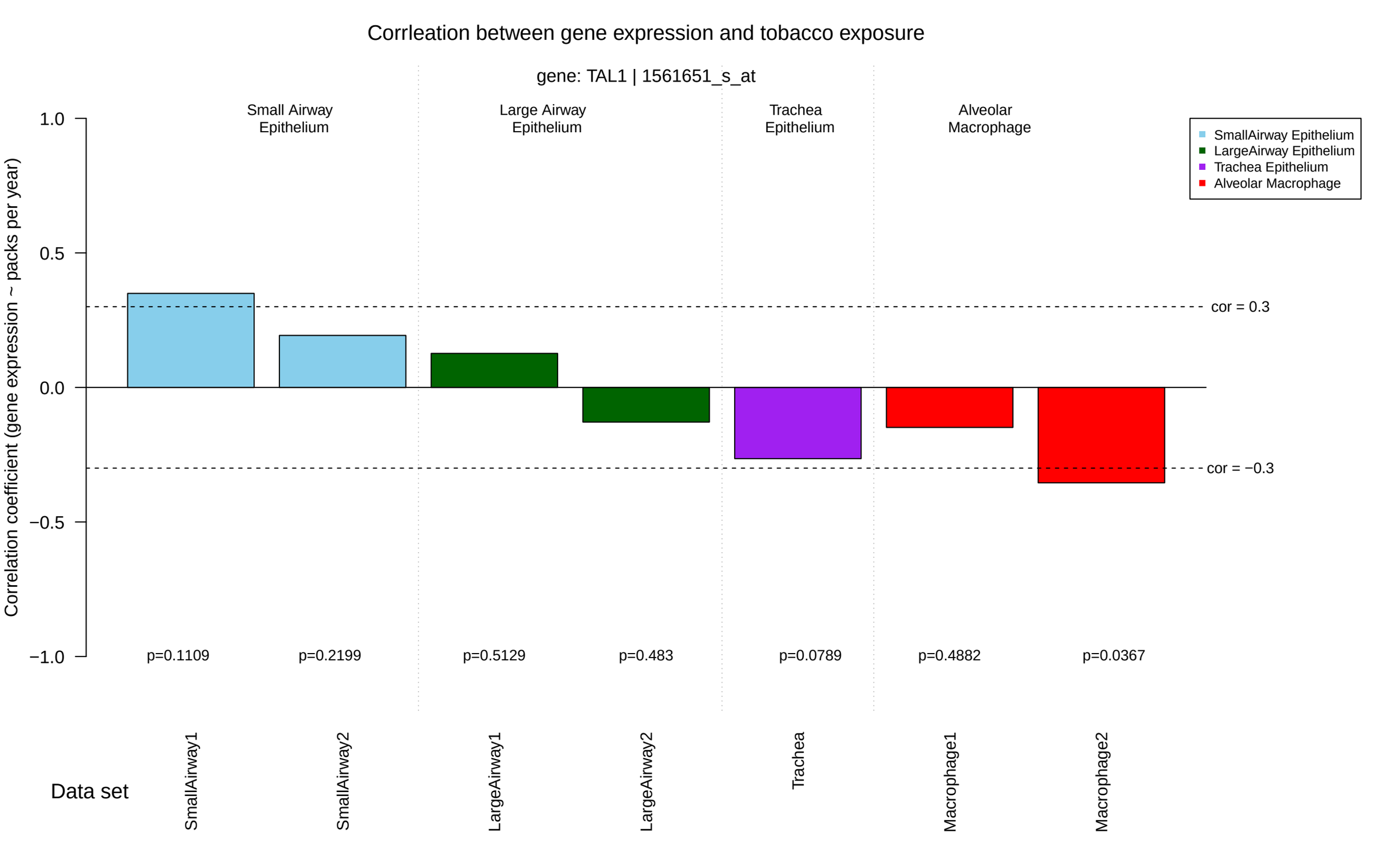

set: LargeAirway1 (p=0.2557)
TAL1 | 1561651_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.7395)
TAL1 | 1561651_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0657)
TAL1 | 1561651_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.9863)
TAL1 | 1561651_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.3292)
TAL1 | 1561651_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: TAL1 | 1561651_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1109
p=0.2199
p=0.5129
p=0.483
p=0.0789
p=0.4882
p=0.0367
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2